DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning

INTERNSHIP REPORT

ON

"Exploring Data Science and Gen AI Solutions: Internship Experience at AT&T"

Submitted in partial fulfillment for the award of degree(*21INT82*)

BACHELOR OF ENGINEERING IN

Department of Artificial Intelligence and Machine Learning

Submitted by:

Amit A (1DS21AI002)

Conducted at

AT&T (American Telephone and Telegraph Company)

Department of Artificial Intelligence and Machine Learning

Dayananda Sagar College of Engineering Bangelore-78

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CERTIFICATE

This is to certify that the Internship titled **“Exploring Data Science and Gen AI Solutions: Internship Experience at AT&T”** carried out by **Amit A [1DS21AI002]**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering**, in **Artificial Intelligence and Machine Learning** during the year 2024-2025. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (21INT82)

______________	______________	______________
Signature of Reporting Head	**Signature of Internship Coordinator**	**Signature of HoD**
Sonu Chandra	**Dr. Vindhya P Malagi**	**Dr. Vindhya P Malagi**
Assoc Director- Data Science	Dept. of AI & ML DSCE	Dept. of AI & ML DSCE

External Viva:

Name of the Examiner	Signature with Date
1) ________________	________________
2) ________________	________________

DAYANANDA SAGAR COLLEGE OF ENGINEERING

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Department of Artificial Intelligence and Machine Learning

D E C L A R A T I O N

I, **Amit A,** final year student of Artificial Intelligence and Machine Learning, Dayananda Sagar College of Engineering, Bangalore - 560 078, declare that the Internship has been successfully completed, in. This report is submitted in partial fulfillment of the requirements for the award of Bachelor's degree in Artificial Intelligence and Machine Learning, during the academic year 2024- 2025.

Date: USN: 1DS21AI002

Place: Bangalore NAME: Amit A

OFFER LETTER

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December 3, 2024

To,

Amit A
Dayananda Sagar College of Engineering

REF: INTERNSHIP AND PRE-PLACEMENT OFFER WITH AT&T COMMUNICATION SERVICES INDIA PVT LTD.

Dear Amit,

With reference to the interview that you have had with us, it gives us great pleasure to offer you an **Internship** at **AT&T Communication Services India Pvt Ltd.** (the "Company"). In addition to confirming the offer, this letter will describe the terms and conditions of your Pre-placement offer. We hope that this opportunity will provide you with exposure to technology and corporate work culture besides helping you to fulfill the criteria for your academic course.

Title: Your title will be **'Staff Associate Technical Intern'**. You will be informed about your manager by the Company at the time of joining and subsequently whenever there is a change in manager.

Terms and Conditions:

Location: You will report for joining at **Bengaluru** office. Post onboarding training you will be required to work from Hyderabad or Chennai office as per business requirement.

Effective Date: Your internship duration will be **Six Months** from **January 7, 2025** or other mutually agreed date and may be extended by mutual consent based on performance appraisals and progress of learning.

Timings: The work hours during the internship will be 8 hours per day besides a meal break of one hour as per the company policy, which may include shift timings.

Payment Terms: This internship is for the purpose of training only and will not be deemed to be employment. You will receive a stipend of **Rs. 49627.00** per month during your period of internship, less all applicable taxes. This stipend is assuming attendance on all the company working days during the month; stipend will be paid pro-rata in the event of absence on any grounds. Payment of stipend will be as per the Company's prevailing practice, in arrears, for the completed month.

COMPLETION CERTIFICATE

ACKNOWLEDGEMENT

This Internship is a result of accumulated guidance, direction and support of several important persons. We take this opportunity to express our gratitude to all who have helped us to complete the Internship.

We express our sincere thanks to our Principal **Dr. B G Prasad**, for providing us adequate facilities to undertake this Internship.

We would like to thank **Dr. Vindhya P Malagi**, Head of Department – AI & ML, for providing us an opportunity to carry out Internship and for her valuable guidance and support.

We would like to thank **Sonu Chandra**, for guiding me during the period of internship.

We express our deep and profound gratitude to our Internship Co-ordinator, **Dr. Vindhya P Malagi**, Head of Department – AI & ML, for her keen interest and encouragement at every step in completing the Internship.

We would like to thank all the faculty members of AI & ML department for the support extended during the course of Internship.

We would like to thank the non-teaching members of the Department of AI & ML, for helping us during the Internship.

Last but not the least, we would like to thank our parents and friends without whose constant help, the completion of Internship would have not been possible.

Amit A [1DS21AI002]

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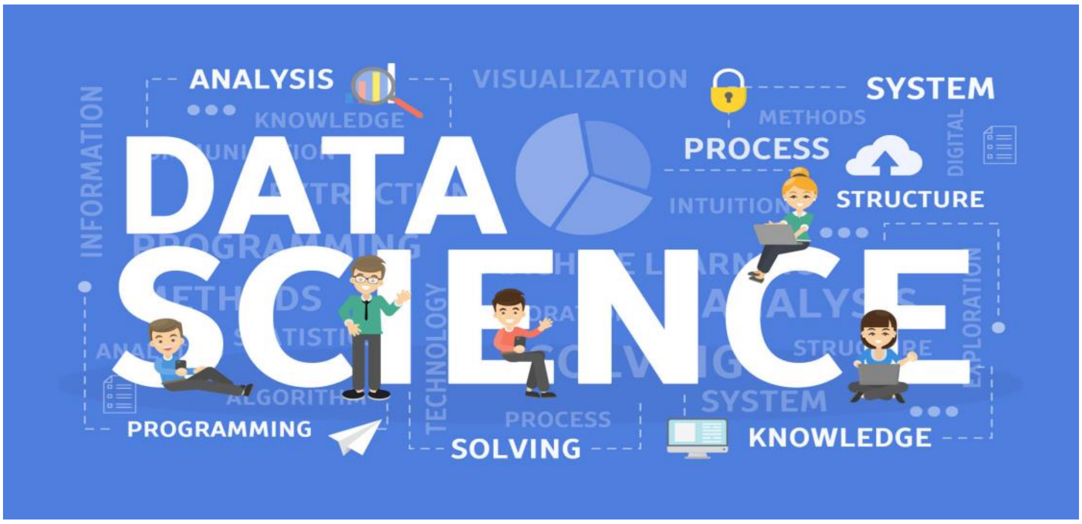
CHAPTER 1
INTRODUCTION

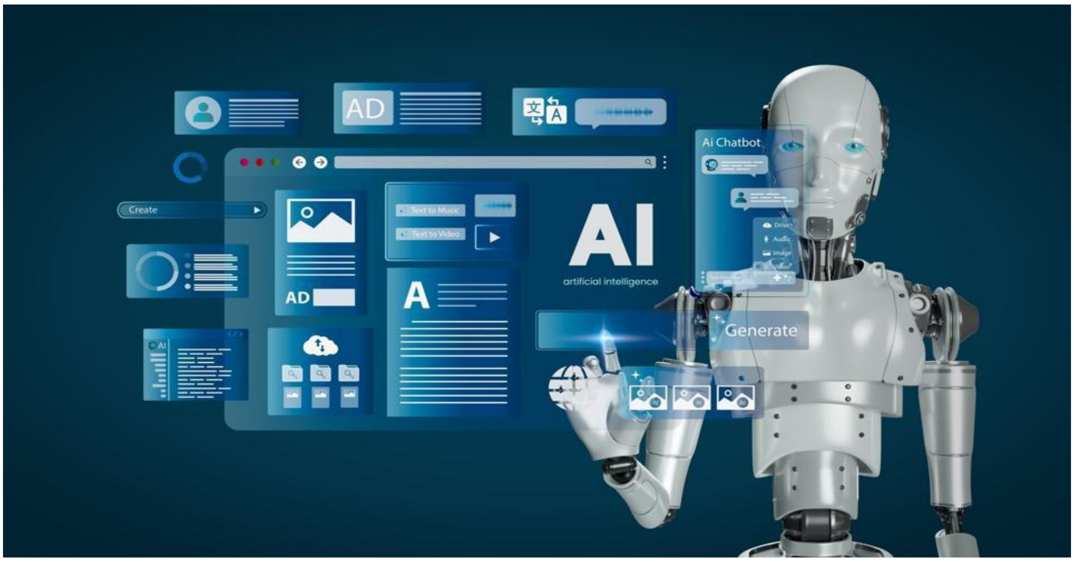
1.1 Introduction to Data Science



Data Science is an interdisciplinary field that uses scientific methods, processes, algorithms, and systems to extract meaningful knowledge and insights from both structured and unstructured data. It combines elements of statistics, computer science, mathematics, and domain-specific knowledge to uncover hidden patterns, trends, and associations in large datasets. As organizations increasingly collect massive volumes of data from various sources—such as social media, sensors, transactions, and customer feedback—the need for data science has grown exponentially. It plays a crucial role in enabling data-driven decision-making, enhancing operational efficiency, and gaining a competitive edge in today's digital landscape.

The typical workflow of data science involves multiple stages, beginning with data collection and preprocessing, followed by exploratory data analysis (EDA), feature engineering, model selection, and evaluation. Data scientists use programming languages like Python and R, along with tools such as SQL for data extraction, and libraries like pandas, NumPy, scikit-learn, and TensorFlow for modeling and analysis. Visualization tools like Matplotlib, Seaborn, and Tableau help communicate results effectively. The success of a data science project depends not only on technical proficiency but also on the ability to interpret results in the context of business goals and communicate them to non-technical stakeholders.

1.2 Introduction to Generative AI



Generative AI (GenAI), a subfield of artificial intelligence, focuses on creating new content by learning patterns from existing data. Unlike traditional AI models that classify or predict based on input data, generative AI models are capable of producing novel data that mimics the characteristics of the training data. Prominent examples include Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), and large language models like OpenAI's GPT (Generative Pretrained Transformer). These models can generate text, images, music, code, and even realistic voices. Applications range from creating chatbots and digital art to automating content creation and simulating complex environments in gaming and virtual reality.

The intersection of data science and generative AI opens up exciting opportunities for innovation. Generative AI can be used to create synthetic data for training machine learning models when real data is scarce or sensitive. It also aids in automating report generation, generating realistic simulations for testing, and improving data augmentation strategies. In turn, data science provides the foundational understanding and tools necessary to evaluate and refine generative models. As both fields continue to evolve, they are transforming industries by enabling smarter automation, deeper personalization, and more intuitive human-machine interactions. The combination of data science and generative AI is shaping the future of how we analyze, generate, and interact with data in an increasingly intelligent world.

1.3 Agentic AI

Agentic AI refers to a class of artificial intelligence systems that are designed to operate as autonomous agents capable of making decisions, setting goals, and taking actions to achieve those goals with minimal or no human intervention. Unlike traditional AI systems that follow predefined instructions or respond to specific inputs, agentic AI possesses a higher degree of independence and adaptability. These systems can perceive their environment, reason about the best course of action, learn from experience, and act proactively. Examples include autonomous robots, personal AI assistants, and AI-powered agents in complex simulations or real-world scenarios like logistics, finance, and customer support.

What sets agentic AI apart is its ability to pursue long-term objectives, dynamically adjust plans, and collaborate with other agents or humans. This involves integrating multiple AI capabilities such as natural language processing, planning, reinforcement learning, and decision-making under uncertainty. With advancements in multi-agent systems and reinforcement learning, agentic AI is becoming more capable of functioning in complex, interactive environments. As the field matures, it raises important questions around ethics, control, and accountability, making responsible design and oversight essential to ensure these autonomous systems align with human values and societal goals.

CHAPTER 2
DESCRIPTION OF ORGANIZATION

AT&T – American Telephone and Telegraph Company

2.1 Introduction

AT&T Inc. is one of the world's largest telecommunications and technology companies, with a rich history that dates back to 1877 when Alexander Graham Bell founded the Bell Telephone Company. Over nearly a century and a half, AT&T has played a pioneering role in shaping global communications, evolving from a regional telephone service provider to a global leader in connectivity, media, and technology. The company provides a wide range of services, including wireless communications, broadband internet, entertainment through streaming and satellite TV, and business services such as cybersecurity and cloud solutions. With its commitment to innovation and transformation, AT&T has been at the forefront of next-generation technologies like 5G, IoT, and edge computing.

Over the decades, AT&T has demonstrated tremendous adaptability and resilience. From its restructuring in 1984 following antitrust actions to acquiring major media companies like DirecTV in 2015 and Time Warner in 2018, the company has strategically expanded into both telecommunications and entertainment sectors. Despite industry challenges and restructuring efforts, AT&T reported annual revenues of over $122 billion in 2023, reflecting steady growth and its ability to respond to changing market dynamics. Today, AT&T continues to enhance customer experience through advanced connectivity solutions, smarter network management, and digital-first innovations across industries.

India plays a vital role in AT&T's global operations. As the first foreign telecom provider to secure ILD and NLD licenses in India, AT&T laid early groundwork to support both multinational and India-based enterprises. Its India Development Center (IDC), located across cities like Bengaluru, Hyderabad, and Chennai, has evolved into a strategic tech and innovation hub. Currently employing over 2,000 professionals—with plans to onboard an additional 2,500–3,000 in the coming years—the IDC supports core engineering, software development, AI/ML projects, network optimization, and next-gen technology initiatives.

With the recent addition of a dedicated digital operations center in Chennai, AT&T continues to strengthen its footprint in the Indian tech ecosystem.

Within AT&T’s global framework, the **Chief Data Office (CDO)** serves as a cornerstone in driving data-centric innovation and enterprise-wide transformation. The CDO division in India is instrumental in AT&T’s data strategy, contributing to data governance, engineering, analytics, and AI model development. By building scalable data platforms, predictive models, and intelligent dashboards, the CDO India team enables faster decision-making, network intelligence, and smarter customer experiences. The division actively supports AT&T’s goals of becoming a data-first organization, ensuring every business unit is equipped with timely, high-quality insights.

As AT&T continues its journey into the future of digital connectivity and intelligent automation, its investments in India—both in terms of talent and infrastructure—are a testament to its long-term commitment. The synergy between global expertise and local innovation, especially through the CDO and IDC, positions AT&T not just as a telecom giant but as a data-driven technology powerhouse shaping the future of global communication.

2.2 Founders

John Stankey – CEO

John Stankey, the Chief Executive Officer of AT&T Inc. since July 2020, has been a driving force behind the company's strategic transformation and renewed focus on core telecommunications services. With a career spanning more than three decades at AT&T, Stankey has held several key leadership roles across media, technology, and operations. Under his leadership, AT&T has sharpened its priorities, advancing nationwide 5G infrastructure, expanding fiber connectivity, and simplifying its business portfolio by divesting non-core assets. His vision centers on making AT&T a connectivity powerhouse that leads with innovation, customer-centricity, and digital-first experiences.

Santosh Bijur – Executive Director and Head of the India Development Center (IDC)

Santosh Bijur, Executive Director and Head of the India Development Center (IDC) at AT&T, leads a dynamic and rapidly expanding team of technology professionals across Bengaluru, Hyderabad, and Chennai. With over 25 years of global IT experience, Bijur is responsible for driving AT&T's innovation and digital transformation from India. His focus includes advanced areas like AI/ML, cloud computing, network intelligence, and agile software development. Under his leadership, the IDC has evolved into a strategic hub contributing to AT&T's global technology roadmap, playing a critical role in delivering scalable, secure, and data-driven solutions that support AT&T's long-term business goals

2.3 CDO – Chief Data Office India

The **Chief Data Office (CDO) division** at AT&T plays a pivotal role in the company's journey toward becoming a data-driven enterprise. As the telecommunications industry rapidly evolves with the integration of AI, 5G, and digital platforms, the CDO division is responsible for managing, governing, and extracting value from AT&T's vast data ecosystem. From enabling intelligent automation to providing actionable business insights, the CDO ensures that data is a strategic asset that powers smarter decisions across the organization. By developing enterprise-wide data platforms, implementing robust data governance practices, and ensuring data quality, the division lays a strong foundation for innovation and operational excellence.

The CDO division also focuses heavily on advanced analytics and AI/ML capabilities. Teams within the division build scalable models for customer segmentation, network optimization, fraud detection, and churn prediction, among many other use cases. These insights not only help improve customer experiences but also drive efficiency across internal operations. The CDO enables business units to transition from reactive reporting to predictive and prescriptive analytics, fostering a culture of innovation backed by data. It partners closely with various departments—technology, marketing, operations, and customer service—to embed intelligence into everyday workflows.

CHAPTER 3

SCOPE OF WORK

During my time at the **Chief Data Office (CDO)** at AT&T, I worked on a diverse array of projects that leveraged cutting-edge technologies to drive data-driven decision-making and enhance operational efficiency. My primary focus areas included **Generative AI**, **Agentic AI**, and **Data Science**, where I contributed significantly to key innovations in **RAG (Retrieval-Augmented Generation)**, **Autogen**, **LangGraph**, and **Machine Learning**.

In the field of **Generative AI**, I specifically worked on **RAG (Retrieval-Augmented Generation)**, which combines retrieval-based models with generative techniques to enhance the accuracy and relevance of responses generated from large datasets. By integrating external data sources with generative models, RAG allowed us to provide contextually rich and precise content, which was crucial for automating content generation and improving AI-based customer support systems. My role was to build a use case for Single PDF and Multiple PDF's RAG with a feature of Summarizing, Q&A on the documents etc. for better understanding and then optimizing the retrieval of the documents to reduce the latency.

I was also heavily involved in **Agentic AI**, particularly in the development of **Autogen** and **LangGraph** use cases. **Autogen** focuses on the creation of autonomous systems capable of self-directed, intelligent decision-making developed by Microsoft.I have used Autogen to learn more on Agentic AI and then craft a monthly newsletter for the CDO division using this flow. **LangGraph**, on the other hand, is a tool that leverages natural language processing (NLP) and graph-based representations to model relationships between entities and derive insights from unstructured data. My contributions are learning LangGraph and to build stateful Agents which are capable to store memory as well.

On the **Data Science** side, I worked extensively with **Machine Learning (ML)** to drive predictive analytics and optimization models. This included developing algorithms for customer segmentation, churn prediction, and network performance optimization. I collaborated closely with cross-functional teams to ensure that the models I built were not only accurate but also aligned with AT&T's strategic objectives. By applying advanced ML techniques, I helped AT&T unlock valuable insights from its vast datasets, driving business outcomes like improved customer retention, more efficient network management, and better-targeted marketing strategies.

In addition to the technical work, I was involved in creating actionable insights through tools like **Power BI**. I developed interactive dashboards and visualizations to help business leaders make data-driven decisions. These dashboards provided real-time insights into key metrics such as customer satisfaction, network performance, and operational efficiency, helping to streamline decision-making processes and improve business outcomes. The integration of predictive models with Power BI allowed AT&T to move from reactive analytics to proactive decision-making.

I also contributed to **ReportLab**, where I focused on automating the generation of dynamic reports. These reports were crucial for tracking the performance of various data initiatives and providing stakeholders with up-to-date insights on KPIs. By automating report generation, we ensured that the data provided was consistent, accurate, and available on-demand, making it easier for teams across the organization to access and act upon critical information.

Overall, my work at the CDO division at AT&T gave me the opportunity to engage with state-of-the-art technologies in **Generative AI**, **Agentic AI**, and **Machine Learning**, contributing to the development of smarter, more automated systems that enhanced both internal operations and customer experiences. The ability to blend cutting-edge AI techniques with data science and visualization tools allowed me to contribute to AT&T's journey toward becoming a more agile, data-driven organization

CHAPTER 4
LEARNING OBJECTIVES

1. Strengthen Knowledge of AI and Machine Learning Models

A primary learning objective during my time at AT&T's Chief Data Office (CDO) was to deepen my understanding and hands-on experience with Generative AI and Machine Learning. I worked on building RAG (Retrieval-Augmented Generation) models, enhancing AI-driven automation processes, and improving customer experience prediction models. I also gained exposure to the development of Agentic AI systems, specifically Autogen and LangGraph, where I contributed to creating autonomous decision-making systems and intelligent language processing models. This hands-on involvement helped me understand how AI models are trained, validated, and deployed in a real-world enterprise environment, emphasizing their potential to enhance both business operations and customer satisfaction.

2. Enhance Data Science and Predictive Analytics Skills

Another key objective was to enhance my skills in data science and predictive analytics. I worked on applying machine learning algorithms to analyze customer behavior, forecast network performance, and predict potential churn scenarios. I used advanced models for customer segmentation and insights, playing a crucial role in improving decision-making processes within AT&T. This experience strengthened my understanding of how machine learning can be leveraged to drive actionable insights from complex datasets and how predictive analytics can provide foresight into business trends, ultimately helping to improve both operational efficiency and customer experience.

3. Improve Proficiency in Data Visualization and Reporting Tools

Throughout the internship, I also focused on data visualization and reporting using tools like Power BI and ReportLab. I created interactive dashboards that helped business leaders visualize key metrics, providing real-time insights into performance indicators such as network health, customer satisfaction, and marketing effectiveness. Additionally, I automated the generation of data-driven reports using ReportLab, which played a key role in streamlining internal reporting workflows. This exposure allowed me to understand how data visualization can support business decisions by making complex data more accessible and actionable.

4. Develop Autonomous AI Systems for Real-Time Decision-Making

A significant learning opportunity was working on Agentic AI projects like Autogen and LangGraph, which are designed to support autonomous decision-making in real time. I contributed to developing systems that can analyze data inputs, make informed decisions without human intervention, and continuously adapt to evolving conditions. This experience gave me valuable insights into the challenges of building AI systems that are capable of handling dynamic, unstructured data and making real-time decisions that align with business needs, particularly in a high-stakes, fast-paced environment like telecommunications.

5 .Collaborate Effectively in Agile Development Environments

Finally, the internship gave me a valuable opportunity to work in an agile development environment, collaborating with cross-functional teams on real-time projects. I participated in sprint planning, code reviews, and troubleshooting during the implementation of AI and data science models. Using version control systems like GitLab, I managed code updates, handled merge requests, and participated in feature implementation. This experience taught me how to effectively communicate technical information within a team, manage development cycles, and ensure that projects are delivered on time while maintaining high standards of code quality and system performance.

CHAPTER 5
CHALLENGES AND SOULTION

1. Manual Newsletter Creation Challenge and Solution Through Agentic AI Automation at AT&T Chief Data Office

Challenge:

At the Chief Data Office (CDO) division of AT&T, the monthly internal newsletter—intended to highlight project updates, research milestones, team achievements, and upcoming initiatives—was traditionally created through a manual process. This approach posed several challenges: it was time-consuming, inconsistent, and often error-prone. Gathering updates required coordination across multiple teams, involving emails, calls, and shared documents, which led to delays and missing information. The lack of a standardized structure meant every edition varied in tone and format, which affected readability and engagement. Additionally, the manual effort involved in editing, formatting, and distributing the newsletter consumed valuable time that could have been spent on higher-impact analytical tasks. As the volume of updates and projects grew, the manual method became increasingly inefficient and unsustainable, making it clear that a scalable, intelligent solution was necessary.

Solution:

To address this, we implemented an automated newsletter generation system using Agentic AI, specifically leveraging the Autogen framework to orchestrate a team of collaborative AI agents. These agents worked together to perform specialized roles: gathering content from internal sources (emails, dashboards, documents), summarizing it into concise updates using NLP techniques, applying a consistent tone and structure, and compiling the final newsletter in a polished format. The entire pipeline was coordinated through a LangGraph-like architecture, allowing stateful interaction and memory between agents for continuity and coherence. The system significantly reduced the manual workload, ensured no critical update was missed, and produced consistent, high-quality newsletters in a fraction of the time. Moreover, it laid the groundwork for future enhancements like role-specific personalization and real-time distribution, transforming what was once a repetitive task into a smart, scalable solution that aligned with AT&T's vision of becoming an AI-driven organization.

2. Automating Frontend Development with Vision and Code Generation Models

Challenge:

Frontend development, particularly for beginners and non-technical stakeholders, often involves translating static UI designs or hand-drawn wireframes into actual code—a process that can be tedious, time-consuming, and error-prone. Designers or team members would typically sketch ideas on paper or create mockups in tools like Figma, and then communicate their vision to developers for implementation. This workflow not only introduced communication gaps and delays but also placed the entire burden of coding, styling, and responsiveness on frontend engineers. Additionally, interpreting design intent, accurately placing elements, and maintaining consistency with design standards required experience and deep familiarity with frontend frameworks like React, HTML/CSS, or Tailwind.

Solution:

To solve this, we developed an intelligent frontend coding assistant app powered by two synergistic AI models: a vision model and a code generation model. The vision model was used to interpret screenshots, sketches, or UI mockups, identifying UI components such as buttons, text fields, headers, and layout structures. Once the layout and components were recognized, the coding model—fine-tuned on frontend development languages—automatically generated clean, modular code using technologies like React and Tailwind CSS. Users could upload an image of a UI, and within moments receive the corresponding code, which could be edited or deployed instantly. This dramatically reduced development time, empowered non-developers to turn ideas into prototypes, and accelerated the design-to-deployment lifecycle. The integration of both models into a seamless app interface transformed frontend development into a more visual, accessible, and efficient process.

3. Streamlining Report Generation with ReportLab Automation

Challenge:

Generating PDF reports manually was a repetitive and resource-intensive task that involved collecting data from multiple sources, formatting it in tools like Word or Excel, and then exporting it as a PDF. This process was not only time-consuming but also prone to human error, especially when updates were frequent or when stakeholders requested custom formats and data views. Each time a report needed to be created or modified, team members had to manually adjust tables, align charts, and maintain consistent styles, which hindered productivity and led to inconsistencies across versions. Moreover, the manual approach lacked scalability—generating personalized or dynamic reports for different teams or KPIs became increasingly unmanageable as the volume of data and reporting frequency grew.

Solution:

To address these inefficiencies, we automated the entire report generation workflow using ReportLab, a powerful Python library for creating dynamic PDFs. This solution enabled us to programmatically design and generate well-structured, visually consistent reports directly from our data pipelines. With ReportLab, we built templates that automatically populated charts, tables, and key metrics based on real-time data, removing the need for manual formatting. The reports could be customized per audience, scheduled for delivery, and generated on-demand with zero manual input. This automation not only saved significant time but also ensured accuracy, consistency, and scalability in reporting. By turning a manual bottleneck into a seamless, automated process, we empowered teams with up-to-date, professional reports that supported faster and more informed decision-making.

4. Enhancing Code Quality through Automated Rating and Refactoring

Challenge:

In earlier development workflows, code was often written and submitted without an objective assessment of its quality, readability, or adherence to best practices. This led to inconsistent coding styles, hidden bugs, and maintainability issues that only surfaced during reviews or later stages of deployment. The lack of a standardized metric for evaluating code meant that suboptimal or poorly structured code could easily make its way into the codebase, affecting team productivity and system reliability. Additionally, manual reviews for every small code change were time-consuming and subjective, which slowed down the development cycle and introduced delays in feature delivery and debugging.

Solution:

To establish a quality-first development environment, we integrated pylint—a static code analysis tool for Python—into the development pipeline to automatically rate each piece of code submitted. A minimum rating threshold of 8.5 was enforced, ensuring that only clean, efficient, and maintainable code progressed through the pipeline. If any submission failed to meet the threshold, a Language Learning Model (LLM) was triggered to automatically refactor the code based on pylint feedback, improving structure, readability, and functionality. This combination of static analysis and LLM-powered refactoring created a feedback loop that not only upheld coding standards but also educated developers by showing improved versions of their own code. The result was a significant uplift in code quality, reduced review burden, and faster, more reliable development cycles.

HARDWARE AND SOFTWARE REQUIREMENTS

Software requirements:

Operating System: Windows 10/11

Development Tools:

- Visual Studio Code
- Git
- Data Bricks

Programming Languages and Frameworks:

Technologies:

- Language: Python
- Frontend: Streamlit
- Tools Used:
    - ReportLab – for generating PDF's
    - LangChain and LangGraph, Autogen – for Gen AI and Agentic Workflows
    - Scikit-Learn, Tensorflow– for machine learning solutions

Backend:

- Language: Python
- Framework: FastAPI (used for building secure, asynchronous REST APIs)

Database: MongoDB Compass (for storing reports and issue metadata)

Runtime and Package Managers:

- Python 3.8+ and pip – For managing backend dependencies and running FastAPI servers.

Hardware requirements:

- Processor: Intel i5 or AMD Ryzen 5 and above
- RAM: Minimum 8 GB (16 GB recommended)
- Storage: At least 256 GB SSD
- Internet: Stable high-speed connection

CHAPTER 6

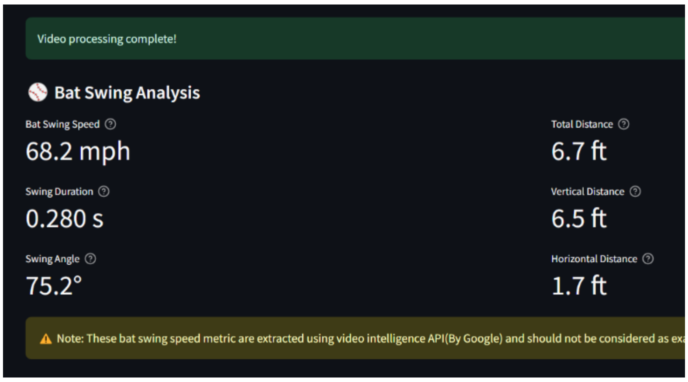
ACHEIVEMENTS AND CONTRIBUTION

During my internship at AT&T, I had the opportunity to contribute significantly to the CDO division at IDC by working on a series of Data Science and Gen AI Use cases, as well as actively participating in development of Agentic workflows. Each contribution not only helped improve my team's performance but also enabled me to strengthen my full-stack development skills.

1. Participated in an Google Hackathon with Team

During my internship at AT&T, I had the incredible opportunity to participate in the Google x MLB Hackathon alongside three of my seniors, forming a passionate four-member team. The hackathon was an exhilarating experience that brought together creative minds from across organizations to solve real-world problems using advanced technologies. Our team focused on a computer vision-based use case, leveraging Google Cloud services like Vision AI, Cloud Functions, and Firebase to build a smart application capable of detecting and analyzing live visual data in real time. It was inspiring to collaborate with experienced peers, brainstorm innovative ideas, and rapidly prototype a working solution under tight deadlines.

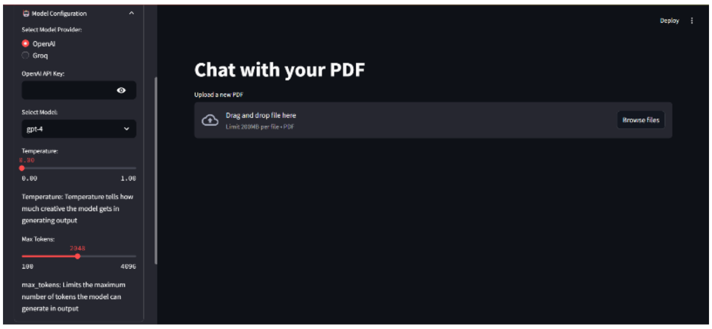
This experience was not only exciting but also immensely fulfilling on both personal and professional levels. I felt a great sense of happiness being part of such a dynamic and supportive team, where learning happened every minute—from understanding new tools to solving complex technical challenges collaboratively. It pushed me out of my comfort zone and gave me hands-on exposure to cloud-driven AI pipelines, something I had only read about before. Participating in the Google x MLB Hackathon enriched my internship journey and significantly boosted my confidence in applying AI technologies to solve meaningful problems.

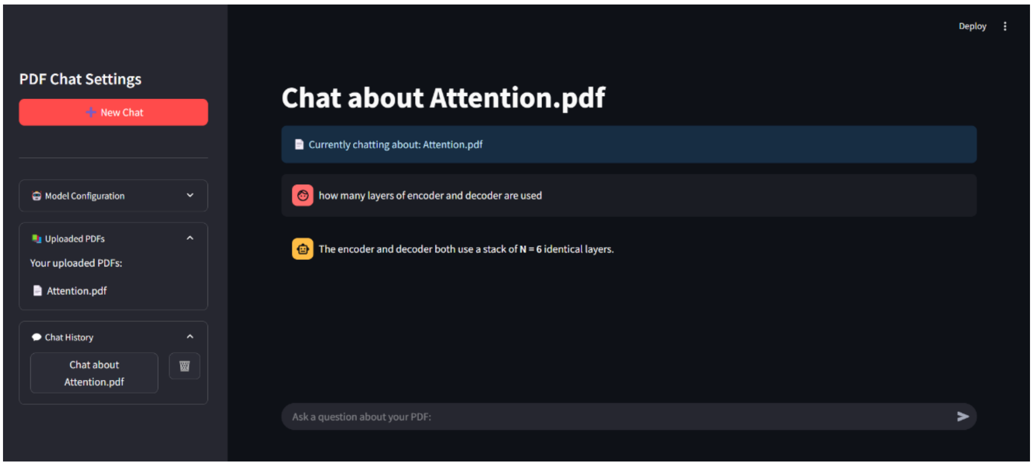


2. Developed Single PDF and Multiple PDF's RAG

During my internship at AT&T's Chief Data Office, one of the most rewarding projects I worked on involved building Retrieval-Augmented Generation (RAG) use cases for both single and multiple PDFs. The challenge was rooted in the manual process of reviewing lengthy documents and extracting relevant information, which was time-consuming and inefficient. We aimed to solve this by developing a solution that could understand, summarize, and answer questions about document content using advanced Generative AI techniques. By integrating RAG pipelines with document loaders, vector databases, and LLMs, we created a system where users could upload a single or multiple PDFs and interact with them through a natural language interface—asking questions, generating summaries, and gaining instant insights.

Working on this project gave me deep exposure to the core architecture of RAG, including document chunking, embedding generation, and retrieval optimization. For multiple PDF handling, we optimized the retrieval process by combining metadata filters and vector similarity searches to reduce latency and improve accuracy. I learned to fine-tune parameters for better relevance, especially when scaling from a single document to multiple sources. It was exciting to see how AI could transform static files into interactive knowledge hubs. This project not only boosted my understanding of LLM-based applications but also showed me how to bridge the gap between unstructured data and meaningful, real-time insights.





3. Multi-Class Classification

During my internship at AT&T's Chief Data Office, I had the opportunity to work on a multi-class classification machine learning task, which aimed to categorize customer behavior into different segments based on usage patterns, service preferences, and network interactions. The challenge stemmed from the complexity of handling large-scale, imbalanced datasets and the need to develop models that could accurately distinguish between multiple customer classes rather than a simple binary outcome. Feature engineering played a critical role, as I had to identify the most impactful attributes while reducing noise from irrelevant or redundant data. I experimented with various algorithms including Random Forest, Gradient Boosting, and Logistic Regression in a one-vs-rest setup, while fine-tuning hyperparameters to improve accuracy and minimize overfitting.

This project was a deep dive into the intricacies of machine learning workflows—from data preprocessing and model training to validation and evaluation using metrics like precision, recall, F1-score, and confusion matrices for each class. I also integrated Power BI dashboards to visualize model predictions and help stakeholders understand customer distribution and model performance in real-time. Collaborating with cross-functional teams gave me insight into business objectives, ensuring our technical solutions aligned with strategic goals. Overall, the experience significantly enhanced my understanding of multi-class classification in a real-world context and helped me appreciate the importance of combining data science with domain knowledge to drive impactful decisions.

Got a better accuracy of around 85% for the task using Gradient Boosting algorithm

4. PDF Search

During my internship at AT&T's Chief Data Office, I worked on developing a PDF search feature that aimed to enhance document retrieval through a more intuitive and efficient search mechanism. The challenge was to build a system capable of searching for specific words, numbers, or phrases within single or multiple PDFs, allowing users to quickly extract the exact data they were looking for. We wanted to go beyond simple keyword searches and enable advanced functionality such as searching for numerical data or specific document sections based on user queries. The solution required integrating advanced text extraction techniques and setting up a robust indexing system to handle large document sets efficiently.

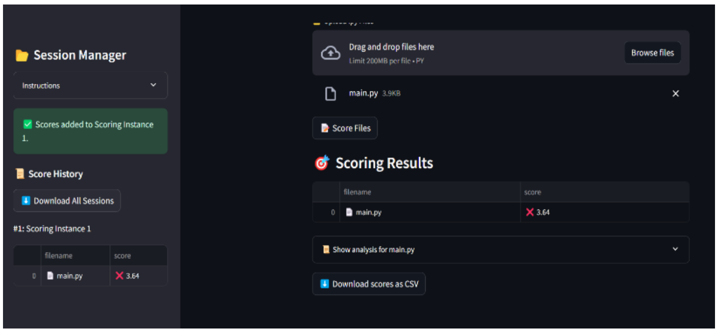
The solution utilized natural language processing (NLP) techniques combined with document parsing to locate precise matches of the desired terms or numbers within the PDFs. We implemented an optimized search algorithm that could handle both exact and fuzzy matching, allowing users to search for exact phrases or numbers, while also accounting for minor variations in formatting or typos. By indexing the extracted content and using vector-based search models, we significantly reduced search time, even when dealing with multiple documents. This project was incredibly rewarding as it helped streamline the process of finding relevant information in large datasets, making document handling more efficient and user-friendly. It also deepened my understanding of document search techniques and their application in real-world systems.

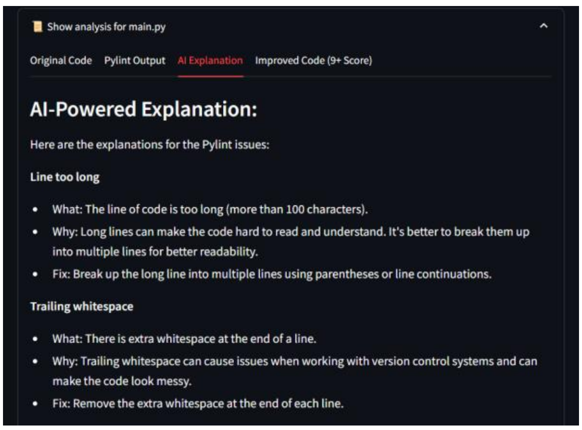
4. Pylint Scoring

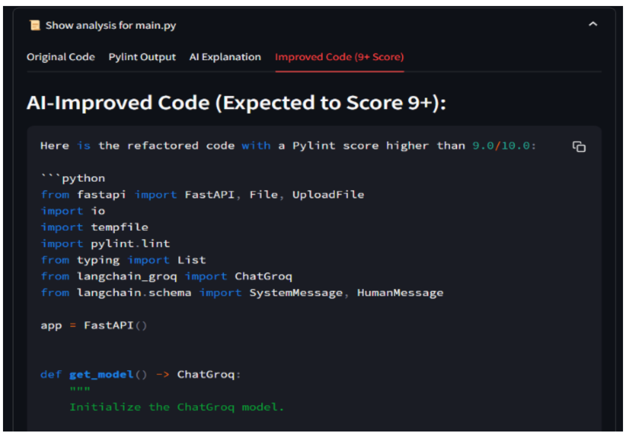
The challenge of ensuring high-quality Python code without manually reviewing each script presented a significant obstacle. Developers often wrote code that initially fell below the required quality threshold, typically scoring below 8.5 on Pylint, a tool that evaluates code based on style, errors, complexity, and other factors. When the score fell below the threshold, it meant the code had to be manually refactored to improve its quality and adherence to coding standards. This process not only took up valuable time but also introduced inefficiencies in the development workflow, as the same code would often be revised multiple times to meet the quality criteria. The lack of an automated system meant that developers spent a disproportionate amount of time fine-tuning their code, rather than focusing on building and improving the core functionality of their applications.

To address this challenge, we integrated **Pylint** into our workflow to automatically assess the quality of the code on every commit. The system was designed to flag any issues with the code, assigning a score that reflected how well it adhered to best practices. If the code's score was less than 8.5, it would trigger an automated process to refactor the code. This was achieved by leveraging a **Large Language Model (LLM)** that could analyze the code, detect common patterns of inefficiency, and automatically refactor sections of the code to improve its structure, readability, and performance. The LLM could make suggestions or directly modify the code to meet quality standards. For instance, it could streamline function definitions, optimize loops, remove redundant code, or adjust naming conventions to enhance clarity and consistency.

By incorporating the LLM into the code quality pipeline, the need for manual code revisions was eliminated. The refactorings performed by the LLM not only improved the overall quality of the codebase but also reduced the workload for developers, allowing them to focus on writing new features instead of spending time improving the code that was already written. Additionally, this system introduced a level of consistency across the project, ensuring that every piece of code adhered to the same set of best practices. With the automated code quality enforcement in place, the development team was able to maintain a high standard of code quality, speed up development cycles, and significantly reduce human error. The use of Pylint combined with the LLM for automated refactoring created a self-sustaining ecosystem where the code continuously improved, driving efficiency and ensuring that the codebase remained clean, efficient, and scalable.







5. Automated PDF generation using ReportLab

The challenge with the manual PDF generation process was that it required constant intervention to ensure that reports were formatted correctly and dynamically adjusted to various content types. Initially, the process was static, with predefined templates and hardcoded structures, which often led to issues such as content overflow, improper image placement, and inconsistencies in the layout. This became especially problematic when dealing with varying amounts of data or documents that had different content structures, such as images, tables, or long paragraphs. The lack of flexibility meant that reports were either incomplete or required significant manual adjustments, which were time-consuming and prone to human error.

To solve this, I created multiple PDF templates using ReportLab, a powerful Python library for generating PDFs. I designed 2-3 different templates to cover various types of reports and their respective content structures, ensuring that each template could handle a specific scenario (e.g., text-heavy reports, reports with images, and those requiring tables). I then worked closely with my senior to help make these templates dynamic by incorporating logic that would adjust the layout based on the content. This included handling edge cases such as content overflow, automatic adjustments to text and image spacing, and ensuring that tables fit within the pages without disrupting the overall layout. I also implemented checks to ensure that images were correctly scaled and aligned within the document, while maintaining a consistent appearance regardless of the content. The automation of the PDF generation process, combined with dynamic templates, greatly improved efficiency, reduced the risk of errors, and ensured that reports were consistently formatted and easy to read.

6. Generated Reports using Power BI

The challenge in generating reports using Power BI was that the process was often time-consuming and required manual intervention at various stages. The data sources were frequently updated, and the reports had to be restructured or modified accordingly. Additionally, there were instances where stakeholders needed customized views or dashboards, which made it difficult to maintain a consistent report structure across different teams. This led to inefficiencies in the workflow, as each report had to be manually adjusted or rebuilt.

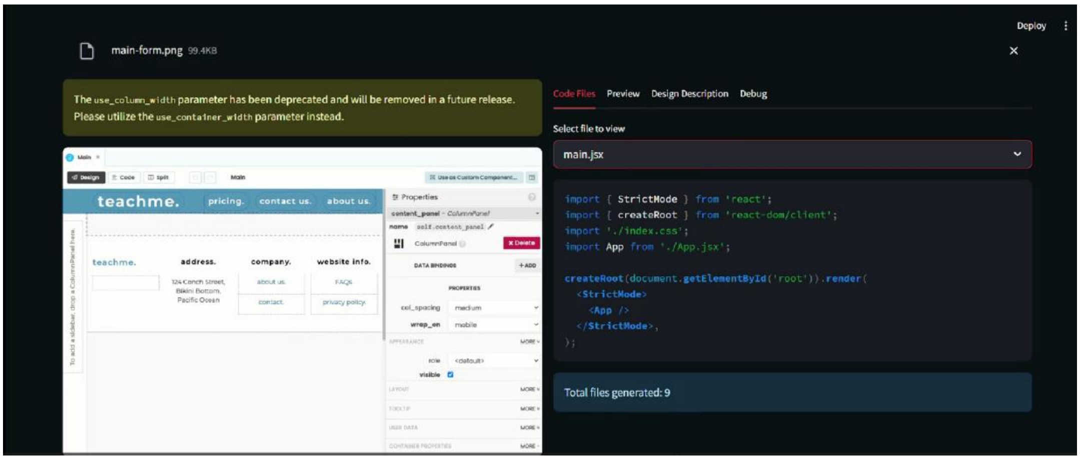
To address these issues, I worked closely with my seniors to streamline the reporting process by leveraging Power BI's advanced features. I helped automate the data refresh process, ensuring that the reports were always up to date with the latest data. I also contributed to designing dynamic and interactive dashboards that allowed stakeholders to filter and explore the data in real time, making it easier to derive insights and make data-driven decisions. By standardizing report templates and incorporating reusable components, we were able to reduce the time spent on report generation and ensure consistency across different reports. Additionally, I assisted in integrating Power BI with other tools and data sources, further enhancing the functionality and interactivity of the reports. My contributions helped increase the efficiency of the reporting process and allowed stakeholders to access real-time, actionable insights, improving decision-making across the organization.

7. UI to Code

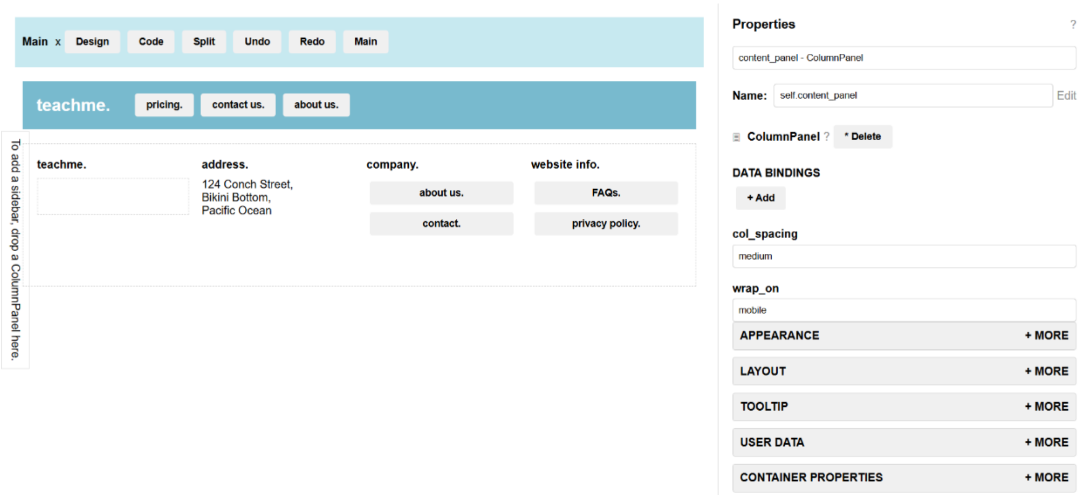
The challenge in developing a UI to code application was that converting designs from Figma or screenshots into functional code required manual effort, and the process often lacked accuracy and consistency. Designers typically provided static assets like Figma designs or screenshots, but translating them into responsive, high-quality code was a time-consuming and error-prone task. Additionally, there were multiple aspects to consider, such as handling various screen sizes, user interactions, and design complexities that were hard to automate using traditional methods. This manual conversion process often led to inconsistencies in the layout, and designers and developers had to go back and forth to adjust the final product, wasting both time and resources.

To address these challenges, I developed a UI to code tool that utilized two models: a vision model and a coder model. The vision model was responsible for analyzing the Figma design or screenshots, extracting layout elements, colors, text, and other design components, while the coder model translated these elements into working code (HTML, CSS, and JavaScript). This integration of computer vision and code generation allowed us to automatically generate functional code based on the design inputs. I also worked on refining the accuracy of these models, ensuring they could handle complex UI components like forms, buttons, and menus. By using these models, the tool was able to produce high-quality, responsive code that adhered to the original design. The automated process drastically reduced the time required to convert a design into code, improved consistency between design and development, and allowed developers to focus on more complex aspects of the project. This solution greatly enhanced the efficiency of the development process, cutting down on errors and unnecessary back-and-forth between teams.

UI of our app:



Code our project gave:



8. Newsletter Automation

Creating the monthly newsletter was initially a manual and time-intensive process. It involved collecting updates from different teams, formatting the content consistently, designing the layout, proofreading for accuracy, and finally compiling everything into a presentable format. This repetitive process often led to delays, version mismatches, and inconsistencies in tone and styling across different editions. The manual effort not only consumed valuable time but also left little room for scalability or personalization, especially when multiple versions were needed for different audiences. Coordinating inputs from various stakeholders further complicated the workflow, leading to potential miscommunication and last-minute edits.

To overcome these challenges, we automated the newsletter generation using agents with the AutoGen framework. Each agent was assigned specific roles—such as content gathering, summarization, layout formatting, and proofreading—allowing parallelized and efficient execution of tasks. The agents could pull data from designated sources, summarize updates intelligently, and format them into a consistent and visually appealing template. Using predefined prompts and dynamic logic, the newsletters were generated in a fraction of the time with improved consistency and accuracy. We also added customization options for different departments or recipients. This automation not only streamlined the entire process but also made it scalable, adaptable, and much more efficient—saving hours of manual effort and ensuring timely delivery every month.

Technology stack

The development of the above use cases involved a modern and scalable technology stack that enabled data science, Gen AI and Agentic frameworks to be developed easily.

Frontend Technologies

- **Streamlit:** A powerful Python-based framework that allowed us to rapidly build interactive and responsive web applications without needing traditional frontend technologies like HTML, CSS, or JavaScript. It enabled us to create clean and functional user interfaces directly in Python code using simple APIs for input widgets, layout components, and data visualization. This significantly reduced development time and made it easy to integrate backend logic seamlessly with the UI, enhancing both developer productivity and user experience**.**

Backend Technologies

- **FastAPI (Python):** FastAPI is a high-performance, modern web framework designed for building RESTful APIs with Python. It enabled rapid development through its intuitive and declarative syntax, making it easy to define endpoints and handle complex routing. One of its key strengths is **automatic request validation** using Pydantic models, which ensured data integrity and reduced boilerplate code. FastAPI also supports **asynchronous programming** out of the box, allowing us to efficiently manage concurrent tasks and improve the scalability of the application. Its built-in **interactive API documentation** (Swagger UI and ReDoc) greatly enhanced the developer experience by allowing real-time testing and exploration of API endpoints.

Database

- **MongoDB Atlas:** A cloud-hosted NoSQL database was leveraged to store user data, issue reports, logs, and other dynamic application content. Its **schema-less design** offered exceptional flexibility, allowing for easy updates and the inclusion of diverse data types without structural constraints. The database supported **real-time synchronization**, enabling instantaneous updates across clients and improving the responsiveness of the application. With **scalable architecture**, it efficiently handled growing volumes of data and concurrent requests without performance degradation. The storage of data in **JSON-like document structures** made it intuitive to map backend models and front-end interfaces, while **built-in security rules and access controls** ensured data privacy and user-specific data management. This solution allowed for rapid development and deployment while maintaining robustness and agility.

API Testing

- **Hoppscotch:** is a lightweight, fast, and open-source API testing tool that runs directly in the browser. Designed as a developer-friendly alternative to Postman, Hoppscotch supports a wide range of protocols including REST, GraphQL, WebSocket, and Server-Sent Events (SSE). It features a clean and minimalistic UI that makes it easy to construct and test API requests, view responses, manage environments, and save collections. With no installation required, it's accessible from anywhere, making it ideal for quick testing and debugging. Hoppscotch also offers features like authentication support, custom headers, history tracking, and even code generation in multiple languages. Its open-source nature allows developers to contribute and extend functionality, making it a flexible and community-driven tool for modern API development workflows.

Version Control

- **Git (GitHub & GitLab):** Version control was managed using Git. GitLab was used during the development phase for branching, staging, and merge requests, while GitHub was used for maintaining the final deployment-ready repository.

Agentic Framework

- **Autogen:** is an advanced framework developed by Microsoft for building multi-agent workflows using large language models (LLMs). It simplifies the process of orchestrating collaborative tasks between AI agents, enabling complex automation pipelines such as code generation, report writing, and decision-making. With built-in support for role-playing agents, customizable communication protocols, and modular design, Autogen allows developers to create scalable and dynamic AI systems. It promotes reusable components and can easily integrate with external tools or APIs, making it ideal for use cases in research, business automation, and AI-driven applications. Autogen empowers developers to design intelligent systems that mimic human-like collaboration and problem-solving, accelerating productivity and innovation.

- **LangGraph:** is a powerful Python framework designed for building stateful, multi-step, and multi-agent workflows powered by language models. It builds on top of LangChain and introduces a graph-based abstraction where nodes represent steps or agents, and edges represent transitions based on logic or model output. LangGraph excels in handling tasks with branching paths, loops, and complex decision-making—making it suitable for chatbots, decision trees, conversational flows, and more. Its flexibility allows developers to orchestrate how LLMs interact with each other or with tools, maintaining context across steps and enabling robust, scalable systems.

LLM's

- **ASKATT: ASKATT** is an advanced orchestration layer built on top of ChatGPT, designed to harness the full potential of GPT models—including GPT-4o—while ensuring data privacy and security. It provides a seamless interface to interact with various GPT versions for diverse tasks such as summarization, coding assistance, content generation, and more. One of its standout features is its robust data protection architecture, ensuring that user data does not leak during model interactions. ASKATT allows developers and organizations to build powerful AI-driven workflows with the flexibility of using multiple GPT variants, all within a secure and controlled environment. Its modular and scalable design makes it ideal for enterprise-level deployments and custom AI solutions that prioritize both performance and privacy.

Development Tools

- **VS Code:** The primary code editor used for writing and organizing code. It offered useful extensions for linting, formatting, and Git integration.
- **Terminal & Git Bash:** Used for command-line operations like running servers, pushing code, and managing version control through Git commands.

CHAPTER 7

REFLECTION AND ANALYSIS

The internship at **AT&T** offered an immersive and transformative experience in the fields of **Data Science**, **Generative AI**, and **Agentic AI**. It provided more than just technical learning—it offered a hands-on opportunity to build intelligent systems, collaborate with cutting-edge technologies, and work closely with advanced LLMs, vision models, and orchestration frameworks. The projects spanned a broad spectrum, from newsletter automation using the AutoGen framework to dynamic PDF generation with ReportLab, each contributing to a deeper understanding of AI's real-world impact and utility.

Throughout the internship, I actively engaged in multiple high-impact projects. I worked on developing a front-end-to-code application that converts UI designs into working code using a **vision model** and a **coding model**. I also built a Streamlit-based UI for users to interact with this system. I explored code quality scoring using **Pylint**, creating a pipeline that uses LLM-based refactoring for code below a certain threshold. In addition, I helped automate newsletter generation using agentic workflows, built single and multiple PDF RAG (Retrieval-Augmented Generation) pipelines, and implemented search systems for PDFs that can identify exact keyword counts and types. I also supported dynamic PDF template development, contributed to Power BI reporting with seniors, and worked on multi-classification ML tasks involving evaluation and model tuning. These initiatives reinforced my skills in Python, data modeling, prompt engineering, and LLM orchestration, while also teaching me the importance of automation, scalability, and intelligent reasoning in modern workflows.

In conclusion, this internship shaped my technical, problem-solving, and collaborative abilities through exposure to cutting-edge developments in AI and data science. It enabled me to not only apply foundational knowledge but to also innovate, iterate, and optimize real-world solutions. From learning how to automate complex tasks using intelligent agents to contributing to scalable AI-driven systems, the journey has been deeply enriching and a crucial step toward becoming a confident and capable AI practitioner.

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CHAPTER 8

RECOMMENDATIONS

1. Introduce Structured AI-Centric Internship Tracks

AT&T can enrich future internship programs by establishing well-structured tracks specifically tailored to domains like **Data Science**, **Generative AI**, and **Agentic AI**. Clearly defined project scopes, milestone-based roadmaps, and regular sprint reviews can help interns navigate complex technologies more effectively while aligning with organizational goals.

2. Strengthen Collaborations with Research Labs and Academia

Partnering with academic institutions and AI research centers can create a pipeline of innovation-driven learning. Organizing guest lectures, AI challenge hackathons, and collaborative model-building workshops can expose interns to emerging trends and practical problem-solving approaches within the industry.

3. Develop a Robust Mentorship Ecosystem

Assigning each intern a mentor with expertise in data science or AI modeling can accelerate learning and build confidence. Regular check-ins, code reviews, and brainstorming sessions can help interns not only grasp technical concepts but also understand how these are deployed at scale within enterprise environments.

4. Promote Ownership Through Innovation-Driven Projects

Encouraging interns to propose, lead, or co-develop innovative AI-based mini-projects can foster creativity and ownership. Whether it's automating tasks using agent frameworks or building RAG pipelines for internal use, such opportunities empower interns to apply their learning in impactful ways while contributing value to the team.

5. Facilitate Cross-Domain Exposure and Knowledge Sharing

Organizing cross-functional sessions with teams working on cloud, cybersecurity, and network intelligence can give interns a broader view of AT&T’s tech landscape. Demos, tech talks, and collaborative retrospectives can deepen understanding of how different verticals synergize to build intelligent, scalable systems.

CHAPTER 9
CONCLUSION

In conclusion, my internship at AT&T provided a deeply enriching experience, allowing me to explore the cutting-edge domains of **Data Science**, **Generative AI**, and **Agentic AI**. It was a transformative journey where I had the opportunity to apply academic knowledge in real-world projects, gain hands-on experience with modern technologies, and collaborate with highly skilled teams. The exposure to the practical applications of AI models, the integration of machine learning with business use cases, and the continuous learning environment played a pivotal role in shaping my technical skills and professional development.

Throughout the internship, I contributed to several impactful projects, including the development of **AI-powered applications** and the **automated generation of reports**. These projects not only deepened my understanding of tools like **FastAPI**, **Streamlit**, and **MongoDB** but also enhanced my ability to work with complex AI frameworks. Additionally, I had the opportunity to integrate **Generative AI** models into real-time systems, gaining firsthand experience of their potential and limitations. I was able to work closely with data pipelines, training models, and optimizing code, which further strengthened my technical expertise.

The hands-on experience with **Agentic AI**—including automating workflows and integrating AI agents into software systems—was a particularly fascinating part of my internship. I contributed to the development of models that can make autonomous decisions, enhancing system efficiency and productivity. These applications not only had a direct impact on business operations but also allowed me to see the real-world potential of AI-driven decision-making tools. The cross-functional collaboration with senior engineers and data scientists further enriched my understanding of the AI lifecycle, from ideation to implementation.

Reflecting on the entire experience, I believe that the internship provided invaluable insights into the intersection of AI, data science, and business applications. It has broadened my perspective, equipping me with the skills necessary to thrive in the rapidly evolving field of AI. The projects I worked on helped me grow both technically and professionally, preparing me for future challenges in the AI space. Overall, this internship was a significant stepping stone in my career, providing me with the knowledge, skills, and network to continue advancing in the tech industry.

CHAPTER 10
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